A Retail Digital Rupiah Prototype on Hyperledger Fabric: Selected Controls and a Reproducible Performance Baseline

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***Abstract*—Bank Indonesia's Project Garuda publicly documents wholesale central bank digital currency. This paper reports a retail Digital Rupiah prototype and a separate controls-study extension on Hyperledger Fabric 2.5. The extension uses four peer MSPs—CentralBankMSP, DistributorMSP, ValidatorMSP, and EndUserMSP— and three Go chaincodes for monetary state, customer controls, and payments. ValidatorMSP has read-only audit access. Server-managed X.509 identities distinguish the signing actor, customer subject, and wallet custodian.**

Synthetic unit tests and a chaincode integration smoke test pass the high-risk KYC flow with distinct reviewer and senior approver, restriction and evidence-based release, reserve funding, retail payment, ValidatorMSP reads, and expected endorsement failures. These tests verify the selected scenarios; they do not establish identity authenticity, screening effectiveness, actual PPATK reporting, or institutional compliance.

Performance evidence is secondary and comes from a one-worker Hyperledger Caliper run on the four-organization controls-study extension. Findings are limited to synthetic functional tests and single-host performance measurements.

***Index Terms*—Digital Rupiah, retail CBDC, KYC and AML/CFT controls, endorsement, Hyperledger Fabric, two-tier distribution, performance baseline**

I. Introduction

Central banks are evaluating Central Bank Digital Currency (CBDC) as a digital form of sovereign money [1]. Bank Indonesia pursues CBDC through Project Garuda. Its publicly documented proof of concept targets the *wholesale* tier, specifically a cash ledger for issuance and redemption among the central bank and large financial institutions [2], [3]. The Garuda White Paper also describes a retail tier in which intermediary banks and payment service providers (PJP) distribute Digital Rupiah to households and merchants [3]. That description remains at the policy and architecture level, and we are not aware of an openly available, testable system that implements Bank Indonesia's retail money business processes end to end on a blockchain ledger.

Central-bank digital money involves a sequence of business processes, not a single ledger operation. The central bank *issues* money into its Treasury and *distributes* liquidity directly to a distributor bank or PJP custodian. Custodians serve customers and merchants in the retail tier. Holders *transact*, custodians *redeem* money to the central bank, and the supervisor *reconciles* and audits its circulation. A blockchain implementation must express these processes as deterministic transactions that each organization can validate independently. It must also preserve two-tier intermediation to reduce bank disintermediation [4] and give the supervisor an audit surface.

Most technical studies isolate one part of this problem: permissioned CBDC architecture [5], privacy [6], security threat models [7], or throughput in account-based systems [8]. Deployed retail systems, including Nigeria's eNaira [9], China's e-CNY [10], and the Bahamian Sand Dollar [11], publish design papers but not reproducible performance benchmarks. The surveyed literature therefore provides no integrated, openly measurable retail reference for the Indonesian two-tier model [12], [13].

The primary evaluation in this work tests whether selected customer controls and organizational endorsement boundaries behave as designed in synthetic scenarios. The Caliper results provide secondary performance evidence for the controls-study extension, reported in its subsection.

To address this gap, we implement a Digital Rupiah prototype on a permissioned Hyperledger Fabric network [14]. The prototype maps Bank Indonesia's business processes to chaincode transactions. This paper contributes:

1) A separate controls-study extension with four peer MSPs and three domain chaincodes for monetary operations, customer controls, and payments. Synthetic integration scenarios exercise distinct senior approval, restriction evidence, reserve funding, read-only validation, and the combined endorsement requirements.
2) A two-tier Digital Rupiah chaincode that integrates hierarchical issuance and distribution, retail transfers, and supervisor-facing reconciliation. This extends the wholesale-only Garuda proof of concept with a testable retail implementation.

3) A deterministic transfer protocol with explicit authorization and balance checks, Fabric multi-version concurrency control (MVCC), and atomic wallet updates. The protocol lets endorsing organizations apply the same state-transition rules independently.
4) An open, reproducible Hyperledger Caliper [15] benchmark for the four-organization controls-study extension. It documents the testbed, the 31 workloads, and the functional checks for authorization, validation, and action outcomes. It reports read and write throughput while keeping the single-host, one-worker scope explicit.

II. Background and Related Work

A. Two-Tier Retail CBDC

In the two-tier (intermediated) model the central bank issues CBDC and remains the ultimate liability holder, while regulated intermediaries serve end users and operate customer-facing rails [1], [4]. This preserves the existing division of labor and limits the central bank's exposure to retail operations. Bank Indonesia adopts this structure for Digital Rupiah, with banks and PJP acting as distributors between the central bank and the public [3].

B. Permissioned Blockchain and Hyperledger Fabric

Hyperledger Fabric is a modular, permissioned blockchain with pluggable ordering, channel-based data isolation, and separate execution, ordering, and validation stages [14]. Membership Service Providers (MSPs) govern identity, the crash-fault-tolerant Raft protocol orders transactions, and CouchDB can provide a queryable world state. Fabric has consequently been used in CBDC and regulated-finance prototypes [5], [16]. Hyperledger Caliper [15], [17] measures throughput and latency under different workloads.

C. Authorization and Supervision

Retail CBDC systems require explicit authorization boundaries, safe balance updates, and supervisory visibility. Token-based retail designs also face denial-of-service and double-spend risks [18], so transfer validation must be deterministic. Our chaincode combines organization-aware authorization, wallet-state checks, and supervision events, complementing studies that address privacy [6] or security [7] separately. The closest published prototype is Project Aurum, a full-stack two-tier retail CBDC built for the Hong Kong Monetary Authority [19], which pairs a wholesale interbank system with a retail e-wallet system and, like this work, keeps the two tiers architecturally distinct. Aurum reports no reproducible throughput benchmark, and its source code and technical manuals are released only to BIS member central banks rather than publicly. What distinguishes this prototype among the representative systems in Table I is that its two-tier intermediation and authorization and supervision controls are realized in a publicly available implementation with a reproducible benchmark, rather than described in design documentation alone.

TABLE I
Comparison with Representative Retail CBDC Systems

System	Platform	2-tier	Open	Bench.
e-CNY [10]	Permissioned	✓	×	×
eNaira [9]	Fabric	✓	×	×
Sand Dollar [11]	Permissioned	✓	×	×
OpenCBDC [8]	Account	×	✓	✓
This work	Fabric 2.5	✓	✓	✓

Open: implementation publicly available. Bench.: open, reproducible throughput/latency benchmark. OpenCBDC is account-based and not distributed-ledger-based. Rows span deployed national CBDCs and a research platform, and summarize public information rather than a controlled comparison.

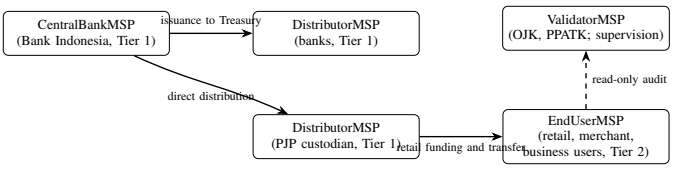


Fig. 1. Two-tier actor structure of the Digital Rupiah prototype, mapped to the four Fabric MSPs. CentralBankMSP (Bank Indonesia) issues Digital Rupiah into Treasury and distributes it directly to a distributor bank or PJP custodian (DistributorMSP), which serves retail, merchant, and business users (EndUserMSP). ValidatorMSP (OJK and PPATK) holds read-only supervisory visibility, shown as a dashed flow.

TABLE II
Bank Indonesia Business Processes and Their Chaincode Realization

Process	Initiator	Chaincode transaction
Issuance	CentralBankMSP	`Mint`, `RequestIssuance`
Distribution	CentralBankMSP	`DistributeToParticipant`
Wallet creation	DistributorMSP	`CreateWallet`
Payment	EndUserMSP	`Transfer`
Redemption	DistributorMSP ValidatorMSP	`Burn`, `RequestRedemption`
Supervision	CentralBankMSP	`GetReconciliationReport`, `GetMetrics`, `GetSupervisionEvents`

ValidatorMSP (OJK, PPATK) has read-only audit access and is not an ordinary payment endorser.

III. Bank Indonesia Business Processes

The system implements six processes in the circulation of central-bank digital money. Fig. 1 shows the actors in each tier. Table II identifies the initiator, chaincode transaction, and endorsement requirement for each process.

The institutional tier contains CentralBankMSP (Bank Indonesia), distributor banks, and PJP custodians (DistributorMSP). These participants follow a lifecycle of submission, approval, freezing, unfreezing, rejection, and offboarding. Bank Indonesia mints Digital Rupiah only to the Treasury wallet, then distributes liquidity directly to an active bank or PJP through `DistributeToParticipant`. Each custodian serves customers and merchants (EndUserMSP) in the retail tier, where the shared transfer path applies ownership, wallet-status, authorization, and balance checks.

IV. System Architecture

Fig. 2 shows the four-layer architecture: client, backend API, off-chain store, and Fabric network. The web client calls

TABLE III
TOPOLOGY OF THE FOUR-ORGANIZATION FABRIC NETWORK

Organization	Role	Peer	Host port
CentralBankMSP	Issuer; hosts orderer and CA	peer0.centralbank	17051
DistributorMSP	Distributor (banks, PJP)	peer0.distributor	18051
ValidatorMSP	Read-only audit (OJK, PPATK)	peer0.validator	19051
EndUserMSP	End users (retail, merchant, business)	peer0.enduser	20051

Full peer names end in `.rupiah.study`, for example `peer0.centralbank.rupiah.study`. Peers listen on container port 7051. Orderer: `orderer.centralbank.rupiah.study` (single-node Raft, host port 17050). Certificate authority: `ca.centralbank`. One CouchDB per organization.

a Go REST backend built with Gorilla Mux. The backend authenticates requests with JWT and role-based access control, stores account and authentication data in PostgreSQL, and submits ledger transactions through the Fabric Gateway SDK. Fig. 3 details the permissioned network.

A. Network Topology

The four-organization network runs one peer per organization on one application channel, `rupiah-study` (Fig. 3, Table III). `CentralBankMSP` represents Bank Indonesia, the issuer. It also operates the single orderer and hosts the certificate authority `ca.centralbank`. `DistributorMSP` represents the banks and payment service providers (PJP) that distribute Digital Rupiah. `ValidatorMSP` represents the supervisors: the Financial Services Authority (OJK, *Otoritas Jasa Keuangan*) and the Indonesian Financial Transaction Reports and Analysis Center (PPATK, *Pusat Pelaporan dan Analisis Transaksi Keuangan*). It has read-only audit access and is not an ordinary payment endorser of the controls-study chaincodes. `EndUserMSP` represents retail, merchant, and business users. Each organization operates one peer with its own CouchDB world state and MSP. A single Raft orderer orders transactions; this configuration provides ordering but not crash fault tolerance, which requires at least three consenters. The application-channel `Endorsement` and `LifecycleEndorsement` policies require a majority of the four organization `Endorsement` policies, whereas the controls-study chaincodes use the explicit policies listed in Table IV.

B. Controls-Study Extension

The controls-study extension runs on a fresh, isolated Fabric 2.5 network with one channel, `rupiah-study`, and four peer MSPs: `CentralBankMSP`, `DistributorMSP`, `ValidatorMSP`, and `EndUserMSP`. CentralBankMSP also operates the single orderer. ValidatorMSP provides read-only audit access and is not an ordinary payment endorser. The extension uses a separate PostgreSQL volume and synthetic state. Server-managed X.509 signers are mapped per application username; the signing identity, customer subject, and wallet custodian are represented separately. Fig. 4 summarizes this topology and the three chaincode policies.

The implementation separates monetary state, customer controls, and payments into three chaincodes. Table IV shows their default policies and state ownership.

Effective-control keys use a state-based policy requiring CentralBankMSP and DistributorMSP. `FundFromReserve` invokes all three chaincodes in one same-channel transaction, so all three policies apply to that funding flow. These policies and distinct-staff approvals are experimental design choices; they are not presented as statutory endorsement rules.

The compliance benchmark measures the three controls-study chaincodes with one Caliper worker on one host. In run-20261014-01, all 31 workloads are confirmed with three 30-second rounds each, with 31 of 31 preflight checks passing. Thirty of 31 workloads are bracketed by measured probes; the `fund-reserve` upper bound is unresolved because two upper-bound probes failed on fetch errors before measurement. The nearest measured failure is 0.21875 ops/s, about 19% above the confirmed 0.183594 ops/s. All functional checks pass: 20 of 20 scenarios, 207 of 207 authorization cells, 27 of 27 validation cases, and 15 of 15 action outcomes. Read operations sustain 272 to 296 operations per second offered, with all verifications passing; delivered load fell to 0.50 to 0.93 of target in the failing probes, so these are lower bounds on read capacity. Confirmed passing write rates (in ops/s, equal to TPS for writes) range from 0.184 to 4.75 TPS across 21 write workloads, with verified throughput at those rates of 0.200 to 4.613 TPS, the mean of three confirmation rounds per workload; 19 of 21 confirm at 0.78125 ops/s or below. In 20 of the 21 write workloads (fund-reserve excluded, since its failing probe was a fetch error), the first failing probe exceeded the drain limit: verified throughput over duration plus drain must reach 90 percent of the delivered rate, which is equivalent to a drain of at most one-ninth of the probe duration (1.1 s for 10 s probes). The median drain was 1.73 s for failing 10 s probes and 0.06 s for passing ones. The configured two-second batch timeout is a plausible source of this drain latency, but the run records no block contents, so the link is not isolated. A fresh-network run (run-20261015-01) measured read-balance at 212 ops/s, with a failing probe at 224 ops/s, and read-receipt at 236 ops/s, with a failing probe at 248 ops/s. Both are bracketed by measured probes, and the fund-reserve result is inconclusive.

C. On-Chain and Off-Chain Data

The Go chaincode in `digital_rupiah.go` groups functions for participants, wallets, issuance and redemption, retail transfers, and supervision (Table V). Account and authentication data remain in the off-chain PostgreSQL store, while the ledger contains the wallet and transaction state needed for settlement and oversight.

A Go backend enforces role-based access control over six roles, summarized in Table VI, before any transaction reaches the chaincode.

The two layers enforce different things, and the split matters for how the prototype's guarantees should be read. The backend maps JWT roles to permitted operations and restricts customers to their own wallets. The chaincode checks

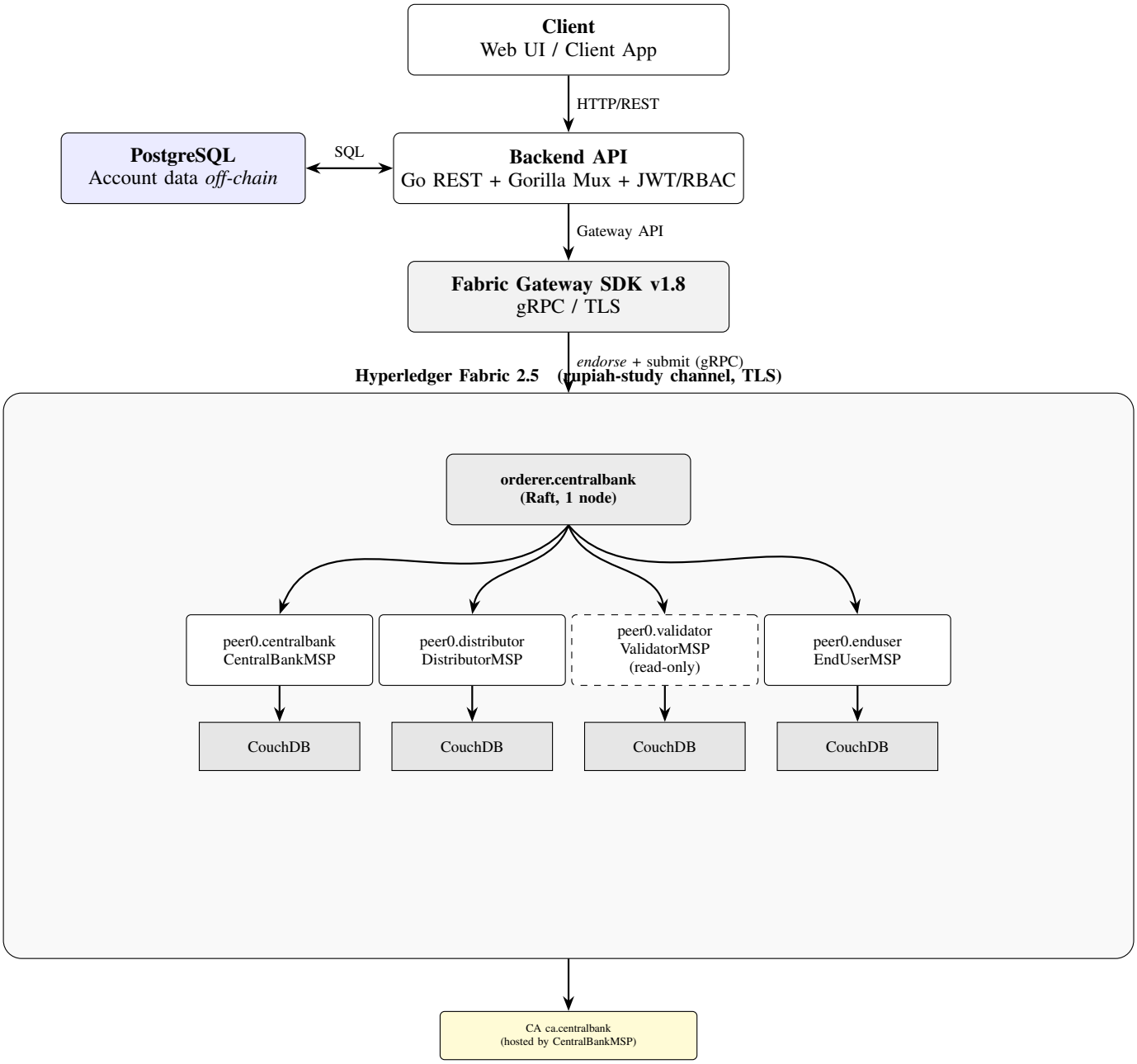


Fig. 2. Architecture of the Digital Rupiah prototype: client layer, Go REST backend with JWT/RBAC, off-chain PostgreSQL store, Fabric Gateway SDK, and the Hyperledger Fabric 2.5 network of four peer organizations on the rupiah-study channel, with one CouchDB per organization, a single certificate authority, and a single-node Raft orderer.

TABLE IV
CONTROLS-STUDY CHAINCODE DOMAINS AND ENDORSEMENT POLICIES

Chaincode	State owned	Default endorsement
`rupiah-monetary`	Treasury, supply, distributor reserves	`AND('CentralBankMSP.peer','DistributorMSP.peer')`
`rupiah-controls`	Pseudonymous subjects, KYC/risk status, business authority, limits, restriction evidence	`AND('DistributorMSP.peer','EndUserMSP.peer')` `OR('CentralBankMSP.peer','EndUserMSP.peer'))`
`rupiah-payments`	Wallets, payment receipts, usage limits	`AND('DistributorMSP.peer','EndUserMSP.peer')`

the caller's organization, wallet status, transfer amount, and available balance before it writes the state transition. Every endorsing peer re-evaluates these rules independently. The single backend and PostgreSQL instance remain trust and

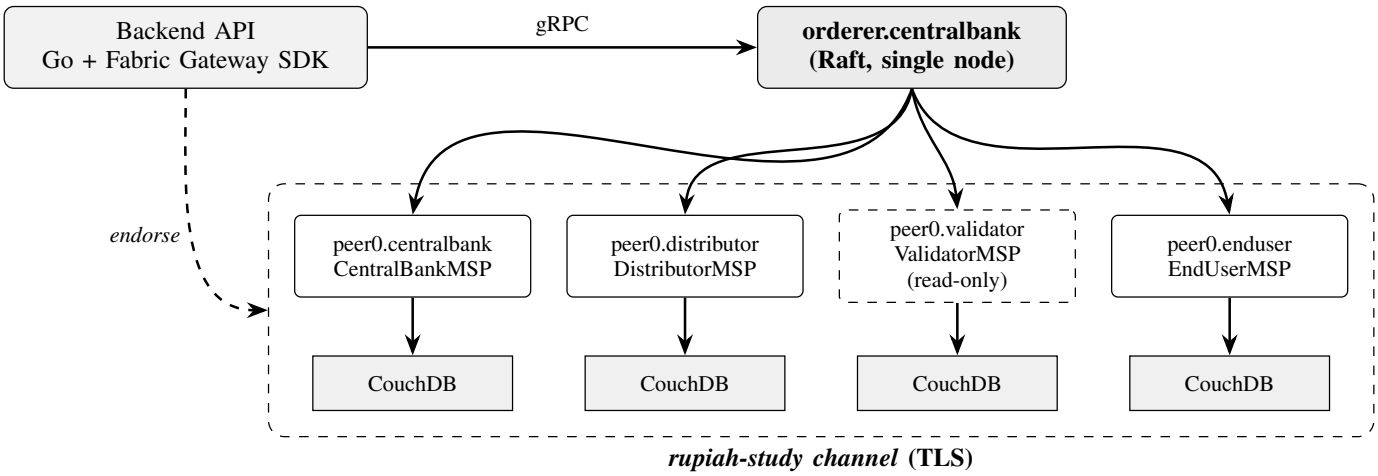


Fig. 3. Hyperledger Fabric topology of the four-organization network: one peer per organization, each with a CouchDB world state, on the rupiah-study channel under a single-node Raft orderer. The Go backend endorses through the Fabric Gateway SDK.

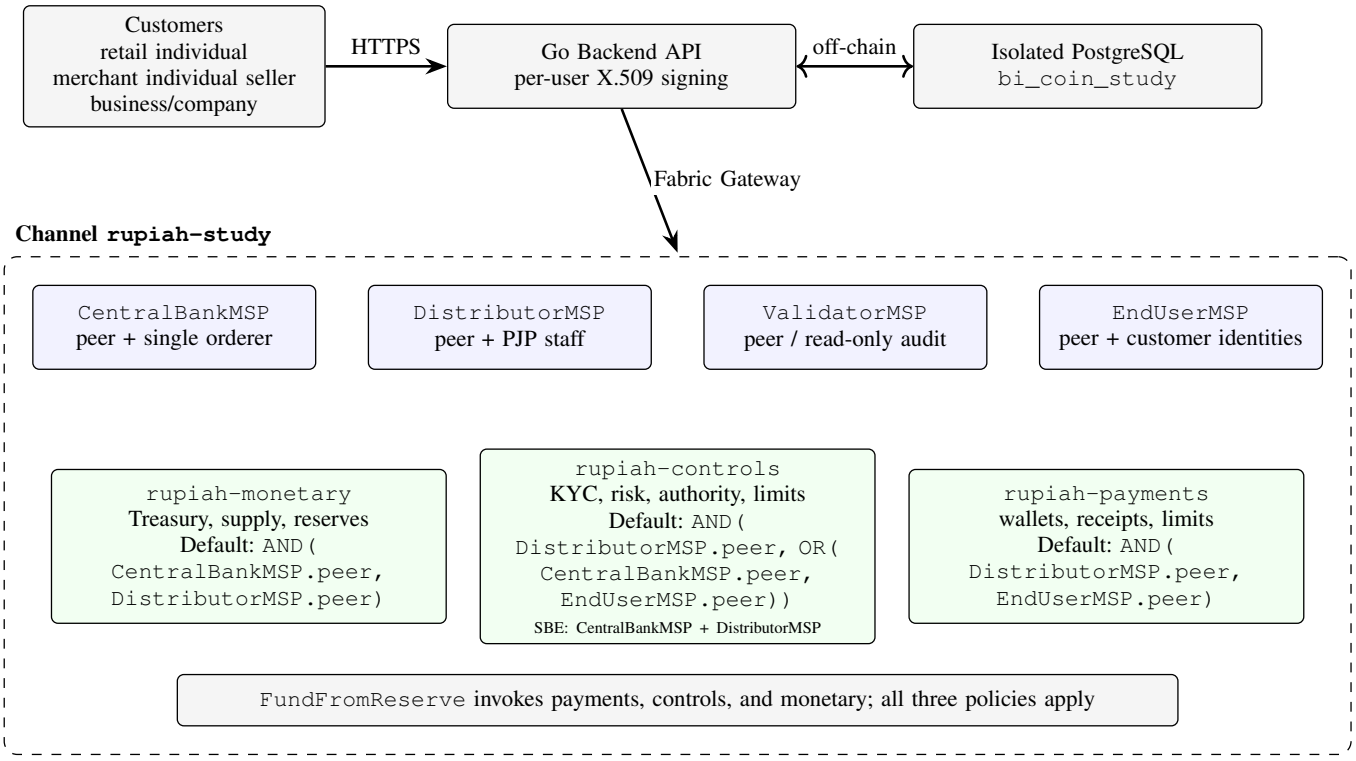


Fig. 4. Four-MSP controls-study architecture, chaincode domains, and endorsement boundaries.

availability concentration points for this prototype.

V. Protocol and Determinism

The transfer path applies explicit authorization and balance checks during endorsement. We formalize these checks and examine their behavior under concurrent transactions.

A. Transfer-Validity Rules

For a transfer of amount a from sender s to receiver r, the chaincode admits the transfer only if every one of the following checks passes:

- the caller's organization and wallet ownership satisfy the authorization rules;
- neither the sender's nor the receiver's wallet is frozen;
- $a > 0$ and the required business reference is present;
- the sender has enough balance to cover a; and
- the sender and receiver wallet records can be updated atomically.

If any check fails, the transfer is rejected. A successful transfer decreases the sender balance, increases the receiver balance, records the transaction, and emits the supervision event used

TABLE V
CHAINCODE FUNCTION GROUPS

Group	Representative functions
Participants	`SubmitParticipant`, `ApproveParticipant`, `Freeze/Unfreeze`, `Offboard`
Wallets	`CreateWallet`, `CreateWholesaleWallet`, `Freeze/UnfreezeWallet`
Issuance	`Mint`, `RequestIssuance`, `RequestIssuanceRtgs`
Redemption	`Burn`, `RequestRedemption`
Distribution	`DistributeToParticipant`
Transfer	`Transfer`
Supervision	`GetReconciliationReport`, `GetMetrics`, `GetSupervisionEvents`

TABLE VI
BACKEND ROLES AND REPRESENTATIVE PERMISSIONS

Role	Representative permitted actions
Public	Health check, login
Authenticated	View own profile, list wallets
Bank or PJP (DistributorMSP)	Create wallet, fund retail wallets
Bank Indonesia (CentralBankMSP)	Issue, redeem liquidity, approve participant
Merchant (EndUserMSP)	Initiate transfer, view balance
Supervisor (ValidatorMSP)	Reconciliation, metrics, supervision events

by the read-only observer path. Treasury distribution accepts only active bank or PJP custodians, and a PJP cannot call issuance directly.

B. Determinism and Double-Spend Safety

Every endorsing peer evaluates the authorization and transfer rules against the same transaction arguments and wallet records. The state transition does not depend on a peer-local clock or on a peer-local cache. Each successful transfer updates the sender and receiver records in one transaction, so every endorser evaluates the same balance transition before validation.

Double-spend safety follows from Fabric's multi-version concurrency control (MVCC). An endorsement reads the current version of each wallet record, and Fabric commits the transaction only if those versions remain unchanged through ordering and validation. Two concurrent transfers from the same wallet may endorse against the same record and version, but only the first ordered transaction commits. The second fails MVCC validation after the first update. Because the balance records are the keys in the read set, one version check covers the state used by the transfer and prevents a stale balance update.

VI. PERFORMANCE EVALUATION

A. Methodology

The controls-study benchmark runs Hyperledger Caliper 0.6.0 with one worker on one host, against the four-organization network and its three chaincodes. Each workload

```
1: function TRANSFER(sender, receiver, amount)
2: if caller unauthorized or either wallet is frozen then
3:     return REJECT
4: end if
5: if amount ≤ 0 or reference is missing then
6:     return REJECT
7: end if
8: if sender.balance < amount then
9:     return REJECT
10: end if
11: sender.balance ← sender.balance − amount
12: receiver.balance ← receiver.balance + amount
13: append transaction history and emit supervision event
14: PutState (validated by Fabric MVCC)
15: return COMMIT
```

Fig. 5. Authorized, atomic, and double-spend-safe transfer executed by the chaincode. Fabric MVCC validates the wallet versions before commit.

TABLE VII
TESTBED CONFIGURATION OF THE CONTROLS-STUDY BENCHMARK

Component	Specification
Host	One host; Intel Core i7-7700HQ, 4 cores, 8 threads, 2.80 GHz
Memory	15.52 GiB
Isolation	Docker-in-Docker
Fabric	2.5
CouchDB	3.3, one instance per organization
Fabric CA	1.5.9, one CA (`ca.centralbank`)
Caliper	0.6.0; one worker; 30 s confirmation windows
Chaincodes	`rupiah-monetary`, `rupiah-controls`, `rupiah-payments`
Channel	`rupiah-study`
Organizations	CentralBankMSP, DistributorMSP, ValidatorMSP, EndUserMSP; one peer each
Ordering	Single-node Raft; batch timeout 2 s; maximum 10 messages

is confirmed over three 30-second rounds, and its verified throughput is the mean of the three rounds. A probe exceeds the drain limit when verified throughput over the probe duration plus drain falls below 90 percent of the delivered rate. The testbed has four peer organizations, one peer per organization, a single-node Raft orderer, and one CouchDB instance per organization.

The Caliper measurements in this section apply only to the four-MSP controls-study extension and its three chaincodes, reported in the Controls-Study Extension subsection.

VII. LIMITATIONS AND FUTURE WORK

The evaluation uses a single host with one peer per organization and no geographic distribution or network latency. Its Caliper results therefore apply only to the reported configuration and do not estimate national-scale performance. Each controls-study workload is confirmed with one worker over three 30-second rounds. A distributed deployment and a graded-skew contention workload would be needed to characterize other operating conditions.

The shared Fabric channel does not provide transaction privacy, and the prototype does not use private data collections or

privacy-preserving cryptography. The single orderer provides ordering but does not tolerate its own failure. The prototype also omits offline payment and leaves connections to domestic infrastructure such as BI-RTGS and BI-FAST as stubs.

The authorization, validation, and action-outcome checks cover the tested paths only. They do not establish formal security, production readiness, or cross-machine reproducibility. Future work should repeat the protocol on distributed hosts, instrument endorsement and commit phases, expand the contention workload, and evaluate privacy, offline payment, and interoperability.

The controls-study extension is limited to synthetic chaincode integration scenarios. It does not use live identity verification or screening feeds, submit reports to PPATK, or establish the effectiveness of a risk model or institutional compliance. Its backend HTTP path is exercised by functional tests but has not been load-measured. Performance results for the three new chaincodes are limited to one Caliper worker on one host with 30-second confirmation windows. The run was interrupted by a host or power interruption and resumed under the same stamp from its last checkpoint. After the restart, the network was brought up again with existing identity material, and chaincode definitions already committed to the channel were skipped. The two payments reads measured after the restart (272 ops/s) are lower than the eight controls and monetary reads measured before it (296 ops/s). A fresh-network run without a restart also measured lower read rates (read-balance 212 ops/s, read-receipt 236 ops/s), so the difference is consistent with run-to-run variation and is not attributed to the restart. The failure classifier used in this run counted transport errors as transaction failures; the fix was made after the run and is not reflected in its results. The current harness stops a search on a transport error as inconclusive, without automatic retry.

VIII. Conclusion

This paper implements Bank Indonesia's digital money business processes in a two-tier Digital Rupiah prototype on a four-organization Hyperledger Fabric network. The chaincode integrates hierarchical issuance and distribution, retail transfers, organization-aware authorization, and supervisor-facing reconciliation.

The transfer protocol combines explicit wallet-state checks with Fabric MVCC. Performance evidence rests on the four-organization controls-study extension.

A separate four-MSP controls-study extension implements three domain chaincodes and passes synthetic integration scenarios for distinct senior approval, restriction evidence, reserve funding, retail payment, read-only ValidatorMSP access, and expected endorsement failures. This functional evidence does not depend on the Caliper results above.

Artifact Availability

The digital-rupiah chaincode, the four-organization network definition, the Go backend, and the Hyperledger Caliper workloads are available at https://github.com/mfachrizalg/bi-coin. The controls-study chaincodes and four-MSP network are under `bi-coin-fabric/study` and `bi-coin-fabric/network/study`, and their Caliper results are reported in the Controls-Study Extension subsection.

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